



HOW CONCEPTION WORKS

The Biology of Conception

Conception is the process by which a sperm cell fertilizes an egg to form a zygote, which then develops into an embryo. Understanding this process can help people make informed decisions about their reproductive health.

Step-by-Step: How Conception Happens

- **Ovulation:** An egg is released from the ovary and travels into the fallopian tube.
- **Fertilization:** If sperm is present, one sperm cell penetrates the egg in the fallopian tube. This is fertilization.
- **Zygote formation:** The fertilized egg (zygote) begins dividing into multiple cells.
- **Blastocyst development:** After several days of division, the cluster of cells forms a blastocyst.
- **Implantation:** The blastocyst travels to the uterus and implants in the uterine lining. This typically occurs 6–12 days after fertilization.
- **hCG production:** Once implanted, the body begins producing human chorionic gonadotropin (hCG), the hormone detected by pregnancy tests.

Factors That Influence Successful Conception

- **Timing:** Intercourse must occur during the fertile window for fertilization to be possible.
- **Sperm quality:** Healthy sperm count, motility, and morphology are all important.
- **Egg quality:** Egg quality declines with age, affecting fertilization potential.
- **Uterine health:** A healthy uterine lining supports successful implantation.
- **Fallopian tube function:** Blocked or damaged tubes can prevent fertilization.

Key Takeaway

Conception is a complex biological process that depends on proper timing, healthy reproductive anatomy, and favorable conditions for fertilization and implantation.

Bloom provides educational guidance only. Always consult with a qualified healthcare professional for personalized medical advice.

Educational content informed by American Society for Reproductive Medicine. (2015). Optimizing natural fertility. Mayo Clinic. (2022). Getting pregnant:

How pregnancy works. Eunice Kennedy Shriver National Institute of Child Health and Human Development. (2022). What is infertility?